Trigonometric Equation: Formula Sheet

Standard equations

$\sin x = 0$: $x = n\pi$
$\cos x = 0$: $x = (2n+1)\frac{\pi}{2}$
$\tan x = 0$: $x = n\pi$
$\sin x = 1$: $x = (4n+1)\frac{\pi}{2}$
$\sin x = -1$: $x = (4n-1)\frac{\pi}{2}$
$\cos x = 1$: $x = 2n\pi$
$\cos x = -1$: $x = (2n+1)\pi$

General solutions

Sine:

$$\sin x = \sin\alpha \Rightarrow x = n\pi + (-1)^n\alpha$$

Cosine:

$$\cos x = \cos\alpha \Rightarrow x = 2n\pi \pm \alpha$$

Tangent:

$$\tan x = \tan\alpha \Rightarrow x = n\pi + \alpha$$

Squares:

$$\sin^2 x = \sin^2\alpha \Rightarrow x = n\pi \pm \alpha$$

Principal α: sin, tan: $[-\frac{\pi}{2}, \frac{\pi}{2}]$; cos: $[0, \pi]$
Reciprocals: cosec, sec, cot as sin, cos, tan

Type 2 and substitutions

$a\sin x + b\cos x = c$: $\sin(x+\theta) = \frac{c}{\sqrt{a^2+b^2}}$, $\tan\theta = \frac{b}{a}$
Solvable: $|c| \le \sqrt{a^2+b^2}$
Half angle:

$$\sin x = \frac{2t}{1+t^2},\ \cos x = \frac{1-t^2}{1+t^2},\ t = \tan\frac{x}{2}$$

$t = \sin x + \cos x$: $\sin 2x = t^2 - 1$
$t = \sin x - \cos x$: $\sin 2x = 1 - t^2$

$$\sin^4 x + \cos^4 x = 1 - \frac{1}{2}\sin^2 2x$$

$$\sin^6 x + \cos^6 x = 1 - \frac{3}{4}\sin^2 2x$$

Transformations (Type 3)

$$\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$$

$$\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$$

$$\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$$

$$4\sin x\sin(\frac{\pi}{3} - x)\sin(\frac{\pi}{3} + x) = \sin 3x$$

$$4\cos x\cos(\frac{\pi}{3} - x)\cos(\frac{\pi}{3} + x) = \cos 3x$$

Bounds (Type 5)

$\sin^m x + \cos^n x \le 1$ $(m, n \ge 2)$

$$-\sqrt{a^2+b^2} \le a\sin x + b\cos x \le \sqrt{a^2+b^2}$$

$$|\tan x + \cot x| \ge 2$$

$2^x + 2^{-x} \ge 2$, $x^2 - 2x + 3 \ge 2$

Inequalities

$\sin x > \frac{1}{2}$:

$$(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6})$$

$\cos x > 0$:

$$(2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{2})$$

$\tan x > 1$:

$$(n\pi + \frac{\pi}{4}, n\pi + \frac{\pi}{2})$$